

Xinhe Zhang

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Summary

AI/ML researcher focused on generative modeling and multimodal representation learning, with experience in efficient generative architectures, large-scale foundation-model adaptation, and agentic AI systems. First author on LiFT and co-first author on two *Science* papers, with hands-on experience in PyTorch, distributed training, large-scale data pipelines, and production software engineering.

Education

Harvard University

Cambridge, MA

Ph.D. Electrical and Computer Engineering (expected Dec 2026)

Carnegie Mellon University

Pittsburgh, PA

M.Sc. Electrical and Computer Engineering (2020) · B.Sc. Electrical and Computer Engineering (2018)

Technical Skills

ML / AI: Generative modeling, multimodal representation learning, foundation models & adaptation, self-supervised learning, vision transformers, parameter-efficient fine-tuning (PEFT; LoRA, adapters), agentic AI (LLMs/VLMs, LangGraph, tool use, RAG), Bayesian optimization

Programming: Python, PyTorch, NumPy, SciPy, scikit-learn, C/C++, MATLAB

Systems & Scale: CUDA, distributed training (FSDP), WebDataset, Linux, Git, Docker, AWS, large-scale data pipelines

Scientific Computing: Computer vision, 3D medical imaging, signal processing, electrophysiology, spike sorting, spatial transcriptomics

Research Experience

Massachusetts General Hospital / Harvard Medical School

Boston, MA

Research Graduate Student, Department of Radiology

Oct 2025–Present

- **LiFT: efficient 3D generative modeling** (*under review at NeurIPS 2026; first author*)

Independently conceived and led the project end-to-end, factorizing 3D image synthesis into high-capacity 2D generation and lightweight inter-slice trajectory modeling. Achieved essentially the same FID as 3D MedDiffusion at ~394x faster inference with 5.5x fewer parameters, and comparable missing-MRI reconstruction quality to cWDM at ~135x faster inference.

- **Large-scale foundation-model adaptation for computational pathology**

Defined and implemented domain adaptation of a ViT-L DINOv3 checkpoint to ~500M H&E slide patches; modified the training pipeline, built a WebDataset input system to address storage-throughput and per-file metadata bottlenecks, and ran/debugged FSDP training on 8x A100 GPUs. Designed downstream evaluation for melanoma relapse prediction and tissue classification; mentoring a student continuing the work.

- **Foundation-model adaptation for prostate cancer imaging** (*with GE Healthcare*)

Built the adaptation and fine-tuning pipeline for an imaging foundation model on multimodal prostate MRI from ~500 patients (T2w, ADC, DWI); evaluating frozen-backbone, full/partial fine-tuning, parameter-efficient adaptation, and multi-task strategies for PI-RADS classification, lesion segmentation, and zonal segmentation.

- **Agentic AI for computational pathology**

Sole developer of a LangGraph-based autonomous literature-mining system spanning 472K+ biomedical articles; orchestrates LLM/VLM agents with a custom deep-learning figure-splitting model to extract and structure multimodal image–text evidence for a computational pathology knowledge base.

Harvard John A. Paulson School of Engineering and Applied Sciences

Cambridge, MA

Graduate Researcher, Ph.D. Electrical and Computer Engineering

Aug 2021–Present

- **Closed-loop AI–bioelectronics for adaptive organoid control** (*bioRxiv 2024; co-first author*)

Built the end-to-end software layer linking 64-channel electrophysiology, waveform processing, Palantir cell-state inference, Bayesian-optimization policies, and programmable stimulation for adaptive cardiac-organoid experiments; AI-guided stimulation increased conduction velocity by ~70% over fixed stimulation. Later built an internal LangGraph/LLM agent supporting autonomous and human-in-the-loop execution of the experimental feedback loop.

- **Multimodal electrophysiology in pancreatic organoids** (*Science 2026; co-first author*)

Built the end-to-end electrophysiology analysis pipeline for 64-channel, 20-kHz recordings, including preprocessing, spike

- sorting, SC- α /SC- β classification, longitudinal maturation analysis, and circadian waveform modeling; analyzed over 2,000 single-cell units and co-developed integration with scRNA-seq and in situ spatial transcriptomics.
- **Cardiac electrophysiology and spatial transcriptomics** (*Science* 2025; co-first author)
Built a custom rule-based signal-processing pipeline to distinguish graft-derived arrhythmogenic automaticity from host sinus rhythm using waveform pattern matching, spatial propagation, and millisecond-scale activation timing; implemented propagation mapping and longitudinal automaticity analysis across 15–135 days and co-developed Slide-seq analysis.
 - **Decoding brain intrinsic dynamics for NeuroAI** (*NIH BRAIN NeuroAI Workshop, poster, 2024*)
Independently designed and implemented a hierarchical generative model in which recurrent long-term context conditions latent within-trial neural trajectory modeling, jointly capturing stimulus-evoked dynamics and long-term representational drift from longitudinal single-cell visual cortex recordings.

Industry Experience

- Duolingo** — Android Developer Pittsburgh, PA | Nov 2018–May 2019
Shipped Android push-notification features for the Chinese market and redesigned purchasing flows, contributing to increased membership sales.
- Facebook** — Software Engineer Intern Seattle, WA | May–Aug 2018
Designed and implemented a full-stack web application for Facebook Analytics Automated Insights, querying databases and presenting results from user-specified analysis parameters.
- Google** — Software Engineer Intern Mountain View, CA | May–Aug 2017
Built an Angular/D3.js interactive visualization system for logical relationships among datasets, used by ~150 monthly business and engineering users; improved filtering and animation workflows used by ~100 daily professional users.
- Google** — Engineering Practicum Intern Boulder, CO | May–Aug 2016
Implemented a C++ backend server retrieving from a large spatially indexed table and built an interactive 3D geospatial visualization using WebGL and the Google Maps API.

Selected Publications

ML / AI Methods

- [1] **X. Zhang**, Y. Zhang, P. Jin, A. Marin-Llobet, N. Li, Q. Li. LiFT: Lifted Inter-slice Feature Trajectories for 3D Image Generation from 2D Generators. Under review at *NeurIPS*, 2026.
- [2] R. Liu[†], Z. Ren[†], **X. Zhang**[†], et al. An AI-Cyborg System for Adaptive Intelligent Modulation of Organoid Maturation. *bioRxiv*, 2024.
- [3] X. Tang[†], J. Zhang[†], Y. He[†], **X. Zhang**, et al. Explainable multi-task learning for multi-modality biological data analysis. *Nature Communications*, 2023.

Biomedical Science

- [4] Q. Li[†], R. Liu[†], Z. Lin[†], **X. Zhang**[†], W. Wang[†], et al. Implanted flexible electronics reveal principles of human islet cell electrical maturation. *Science*, 2026.
- [5] J. Aoyama[†], R. Liu[†], **X. Zhang**[†], et al. Flexible nanoelectronics reveal arrhythmogenesis in transplanted human cardiomyocytes. *Science*, 2025.
- [6] H. Zhao[#], **X. Zhang**[#], A. Marin-Llobet, X. Lin, J. Liu. Benchmarking spike source localization algorithms in high density probes. *PLOS Computational Biology*, 2026.

[#]Co-corresponding author. [†]Equal contribution. Full publication list at xinhez.github.io.

Selected Awards

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| Poster Third Place , Schmidt Center Symposium: Biomedical Science and AI | 2025 |
| Poster Runner-Up & Early-Career Scholar , NIH BRAIN Initiative NeuroAI Workshop | 2024 |

Teaching & Service

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| Teaching Fellow , Harvard (neural computation, neuroengineering, systems development for CS) | 2022–2025 |
| Teaching Assistant , CMU (12 semesters: deep RL, signals & systems, programming) | 2015–2020 |
| Reviewer: <i>Nature Communications</i> , <i>PLOS ONE</i> , NeurIPS/ICML workshops, 10+ journals. | |
| Mentor: First-Year Advisor, Harvard (2022–2024); Academic Advisor, CMU (2020). | |